MICHIGAN DEVFEST HACKATHON 2026 · MANUFACTURING TRACK

ShiftProof

What the work instruction leaves out, reviewed for the next shift.

Josh · Daniel · Alex · Jason

Built on IBM Granite on watsonx.ai · ibm/granite-4-h-small

The instruction says adjust the offset. The veteran says wait.

WORK INSTRUCTION WI-114 REV C · STEP 7

> If the pocket is out of tolerance, adjust the tool wear offset and rerun.

Step 6 says to measure the pocket. It does not say when.

FICTIONAL TRAINING FIXTURE

VETERAN NOTES · MARV K. (FICTIONAL)

> A 4 inch bracket that's maybe 40 degrees over shop temperature will read about two thousandths long.

> I've watched new guys measure it hot, see it over, chase the wear offset, and then every part after that comes out short once it cools.

FICTIONAL TRAINING FIXTURE

An apprentice who follows the sheet to the letter can still make scrap.

Every retirement is a knowledge handoff, and most shops have no system for it

583,000

manufacturing workers in Michigan

Michigan LEO, Oct 2026 [1]

1 in 4

US manufacturing workers are 55 or older (25.2%)

BLS CPS table 18b, 2025 [2]

1 in 3

machine shop workers are 55 or older (31.6%)

BLS CPS table 18b, 2025 [2]

Machinists carry know-how the written work instruction leaves out: how to seat a part, when a measurement can be trusted, where to check twice. When a veteran retires or moves shifts, that knowledge leaves with them.

[1] Michigan Department of Labor and Economic Opportunity, Oct 2026. [2] US Bureau of Labor Statistics, Current Population Survey table 18b, 2025. The age figures are national, not Michigan-specific.

The demo is three clicks: Compare, Review, Ask

1 COMPARE

VETERAN

Granite reads the work instruction beside the veteran's notes. It flags what the instruction leaves out, quotes both sources, and asks a follow-up question. The app checks every quote.

2 REVIEW

VETERAN · QUALITY LEAD

Granite drafts a guide card from the veteran's answer. The veteran confirms it. The quality lead approves it. A card only moves forward.

3 ASK

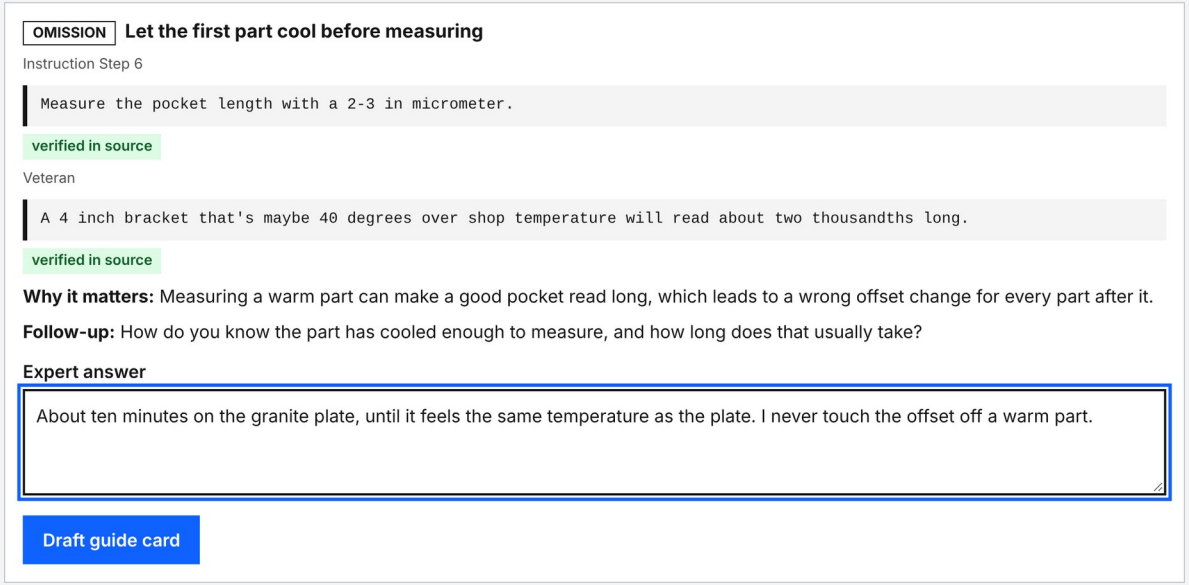
APPRENTICE

At the machine, on a phone. The answer comes only from approved cards and cites the veteran's words. Nothing approved covers it? Ask your lead or quality.

The work instruction itself never changes. ShiftProof adds reviewed context next to it.

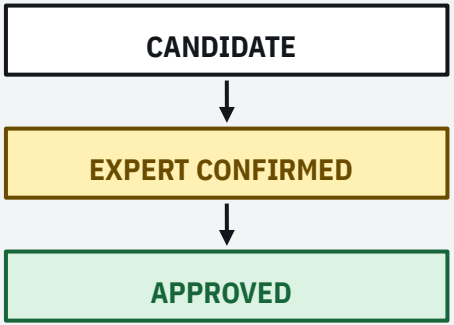
Compare: Granite flags what the instruction leaves out, and the app checks every quote

- WI-114 Rev C beside the veteran's notes, both labelled fictional
- One finding per gap: omission, ambiguity or conflict
- Both quotes copied word for word, and each one verified against its source by the app
- A follow-up the veteran can answer from experience

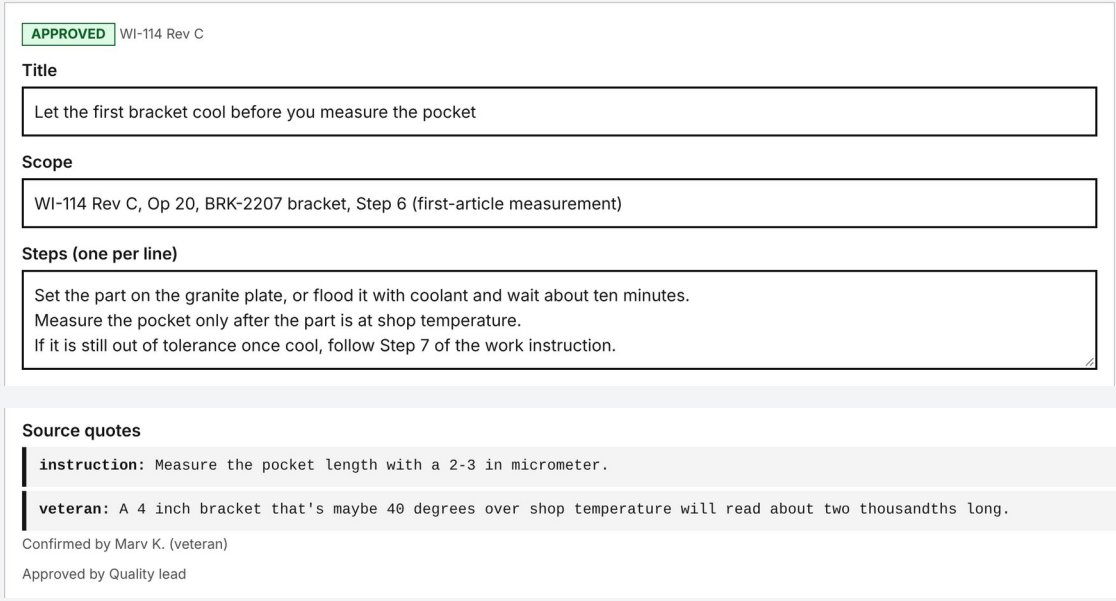


Placeholder capture from the app in offline mode. Replace with the live run from the freeze check.

Review: the veteran confirms, quality approves, and the order can't be skipped



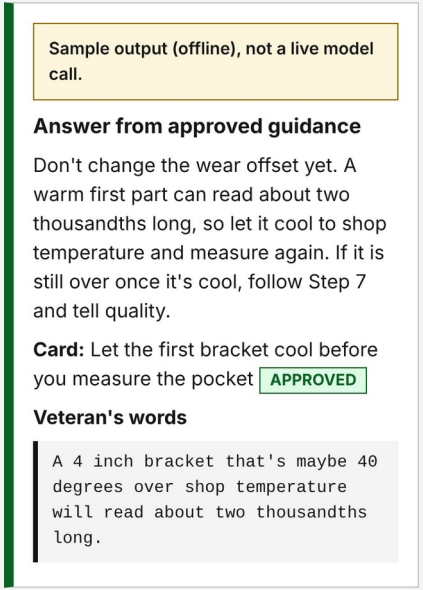
- Granite drafts the card from the veteran's answer
- Any edit sends the card back to candidate
- Skipping a step is rejected by the server



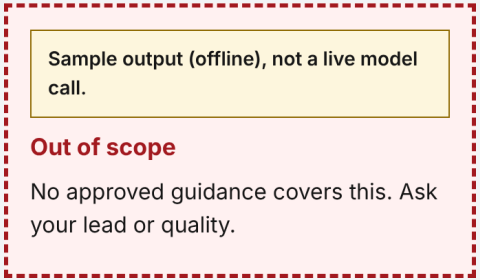
Placeholder capture from the app in offline mode. Replace with the live run from the freeze check.

Ask: the apprentice gets an answer from approved cards only, or a refusal

APPROVED GUIDANCE



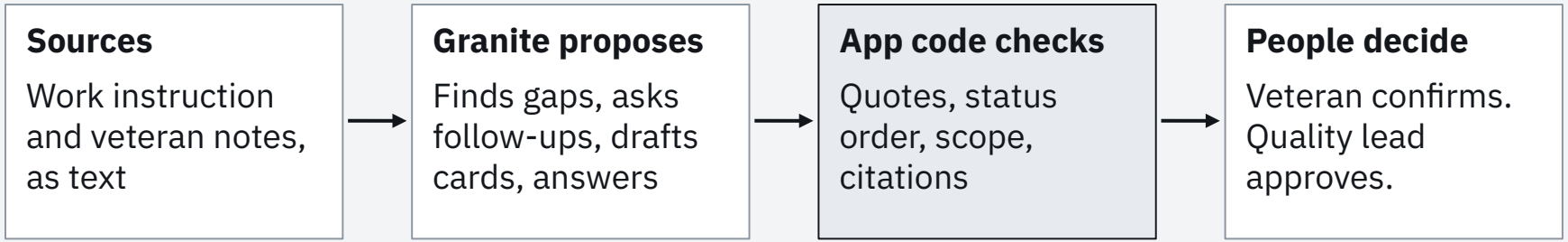
OUT OF SCOPE



Placeholder captures, offline mode, 375 px wide. Replace with the live run.

- The apprentice asks in their own words, on a phone at the machine
- The answer uses only approved cards for WI-114 Rev C and cites the veteran's words
- Nothing approved covers it? The app says so and sends them to a lead. No guesses, no invented numbers.
- No approved card, no model call

Granite proposes. App code decides.



IBM Granite ibm/granite-4-h-small on watsonx.ai · Next.js and TypeScript · model JSON validated with zod

TRUST RULES, ENFORCED IN APP CODE, NOT IN THE PROMPT

1 Every quote is checked against its source. A finding with an unverified quote can't become a card.

2 Cards move one way: candidate, expert confirmed, approved. The server rejects a skipped step.

3 Ask uses only approved cards for WI-114 Rev C. When none exist, the model isn't called at all.

4 Citations are filtered on the server, and the app writes the refusal itself, never the model.

Limits today, and what comes next

LIMITS TODAY

- Demo roles, not real sign-in. The review order is enforced in code; who clicks is a demo switch.
- One procedure, one revision: WI-114 Rev C, a fictional fixture.
- Text in, no audio capture yet.
- Checks pass by hand. Scripted acceptance tests are in progress.

NEXT

- Voice capture, so a veteran can talk through a job at the machine
- Ingest real manuals and work instructions with Docling
- Block reuse when a procedure revision changes
- Move the Ask step and the veteran-to-quality hand-off onto watsonx Orchestrate
- watsonx.governance for audit trails

What the veteran knows, reviewed, at the machine.

SHIFTPROOF · BUILT ON IBM GRANITE ON WATSONX.AI

github.com/critesjosh/mi-devfest

Josh · Daniel · Alex · Jason

AI tools used are listed in the repo README. All demo documents are fictional training fixtures.